# Pastry: routing hops versus network size

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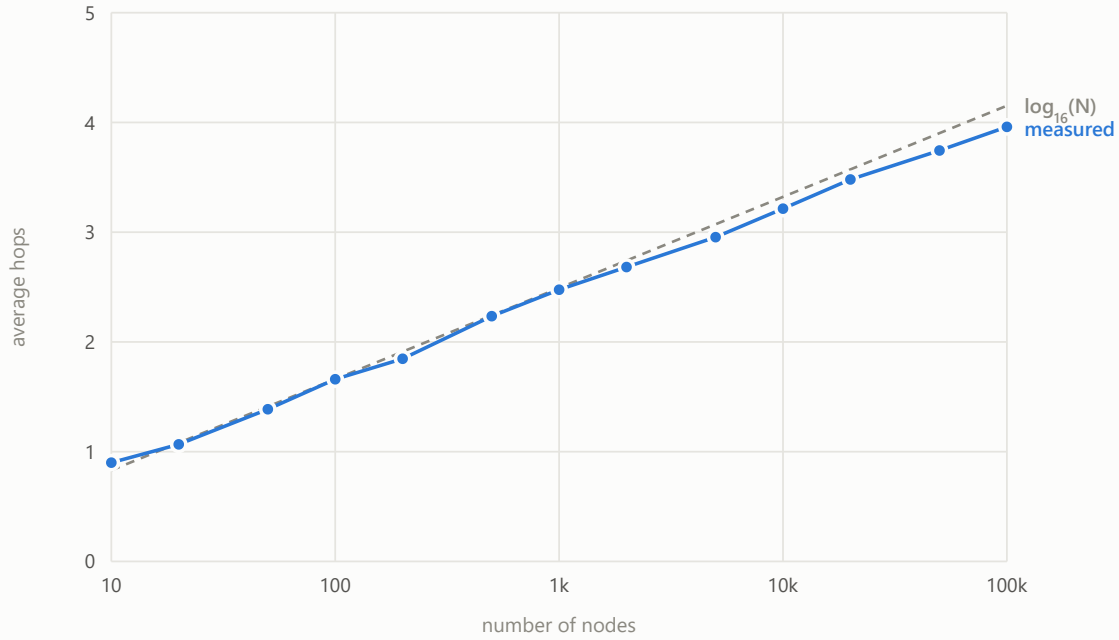
Each peer is an Erlang actor. Every peer makes two lookups for random stored objects, and we count how many hops each lookup takes before it is delivered. Networks of up to 5,000 nodes are averaged over three runs, the larger ones are a single run.

## Average hops

measured $\log_{16}(N)$

### Average hops per lookup

Number of nodes is on a log scale. The dashed line is $\log_{16}N$, the expected number of hops.




## Results

The number of hops grows with log N, which is what Pastry is supposed to give. Since the x axis is on a log scale, this shows up as a straight line. Going from 10 nodes to 100,000 nodes only adds about three hops.

From 100 nodes up, the average is at or a little below $\log_{16}N$ at every size. This makes sense because $\log_{16}N$ assumes we fix one hex digit per hop, but once the key is in range of the leaf set the last few digits are done in a single hop. The Pastry paper reports the same thing.

All lookups found their object, 413,280 out of 413,280 over all runs. We also tested the routing on its own. For each test key we found the closest node by checking every node id, and every lookup ended at that node.

## Largest network

The largest network we ran had 100,000 nodes. 200,000 out of 200,000 lookups were found, with an average of 3.96 hops ($\log_{16}N = 4.15$). On our 8 GB laptop the limit was the time it takes to join the nodes one by one, not memory.

## Measurements

| Nodes | Average hops | $\log_{16}N$ | Found |
|---|---|---|---|
| 10 | 0.900 | 0.830 | 60 / 60 |
| 20 | 1.067 | 1.080 | 120 / 120 |
| 50 | 1.387 | 1.411 | 300 / 300 |
| 100 | 1.660 | 1.661 | 600 / 600 |
| 200 | 1.847 | 1.911 | 1,200 / 1,200 |
| 500 | 2.235 | 2.241 | 3,000 / 3,000 |
| 1,000 | 2.476 | 2.491 | 6,000 / 6,000 |
| 2,000 | 2.682 | 2.741 | 12,000 / 12,000 |
| 5,000 | 2.955 | 3.072 | 30,000 / 30,000 |
| 10,000 | 3.214 | 3.322 | 20,000 / 20,000 |
| 20,000 | 3.480 | 3.572 | 40,000 / 40,000 |
| 50,000 | 3.744 | 3.902 | 100,000 / 100,000 |
| 100,000 | 3.960 | 4.152 | 200,000 / 200,000 |